Fundamentals of Math II **AI Degree**

Recovery Exam **July 5, 2024**

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**Exercise 1.** (0,5 points + 1 point + 2 points+ 0,5 points) Consider the following function $f\colon \mathbb{R}^2 \to \mathbb{R}$:

$$f(x,y) = 10x^2 + 10y^2 - 16xy.$$

(a) Find the gradient $\nabla f(x,y)$ at each point $(x,y) \in \mathbb{R}^2$.

(b) Find all the local extremes of $f$. Use the Hessian to determine whether they are local maxima, local minima, or saddle points.

(c) Use the method of Lagrange multipliers to determine the absolute maxima and minima of $f(x,y)$ subject to $x^2 + y^2 - xy \leq 1$.

(d) Show that the problem $\min f(x,y)$ subject to $x^2 + y^2 - xy \leq 1$ is a convex optimization problem. Mention a theorem given in Theory which implies that the above minimization problem has a unique solution. Is this coherent with your answer in (c)?

**Exercise 2.** (1 point + 1.5 points + 1,5 points) Let $a$ be a positive real number.

(a) Use the change of variable $t = r^2$ to evaluate $\int_0^a e^{-r^2} r dr$.

(b) Use polar coordinates and (a) to compute $\iint_{D_a} e^{-(x^2+y^2)}\, dxdy$, where $D_a$ is the circular disk of radius $a$: $D_a = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq a^2\}$.

(c) Let $S_a$ be the square $S_a = \{(x,y) \in \mathbb{R} : -a \leq x \leq a, -a \leq y \leq a\}$. Show the formula

$$\iint_{S_a} e^{-(x^2+y^2)}\, dxdy = \left(\int_{-a}^{a} e^{-x^2}\, dx\right)^2.$$

Deduce that $\iint_{D_a} e^{-(x^2+y^2)}\, dxdy \leq (\int_{-a}^{a} e^{-x^2}\, dx)^2$.

**Short questions.** (1 point + 1 point)

(a) Determine a parametric curve $r(t) = (r_1(t), r_2(t), r_3(t))$ such that $r(0) = (1,2,3)$ and $r'(t) = (t, \sqrt{t}, te^t)$ for all $t \in \mathbb{R}_+$.

(b) Show that, for $x_1, x_2, \ldots, x_n > 0$, the function

$$f(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i \ln(x_i)$$

is a convex function.

**All answers must be carefully explained. You must specify the theoretical results used in your arguments and procedures.**

SOLUTIONS

① (a) $\nabla f = \begin{bmatrix} 20x - 16y \\ 20y - 16x \end{bmatrix}$.

(b) $\nabla f = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ only one critical point.

$Hf(x,y) = \begin{bmatrix} 20 & -16 \\ -16 & 20 \end{bmatrix}$. This is positive definite, since

$20 > 0$ and $\det \begin{bmatrix} 20 & -16 \\ -16 & 20 \end{bmatrix} = 20^2 - 16^2 > 0$.

Hence $H(f)(0,0) > 0 \Rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is a minimum (local).

(c) Applying Lagrange multipliers, we have to solve:

$$\begin{cases} \begin{pmatrix} 20x - 16y \\ 20y - 16x \end{pmatrix} = \lambda \begin{pmatrix} 2x - y \\ 2y - x \end{pmatrix} \\ x^2 + y^2 - xy = 1 \end{cases}$$

We get: $\dfrac{20x - 16y}{2x - y} = \lambda = \dfrac{20y - 16x}{2y - x}$

Hence $(20x - 16y)(2y - x) = (20y - 16x)(2x - y)$

and thus $x^2 = y^2$. Hence $y = \pm x$.

① $y = x \Rightarrow x^2 + \cancel{x^2} - \cancel{x^2} = 1 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$

We get two points $A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$.

② $y = -x \Rightarrow x^2 + x^2 + x^2 = 1 \Rightarrow 3x^2 = 1 \Rightarrow x = \pm\sqrt{1/3}$

We get two more points: $C = \begin{pmatrix} \sqrt{1/3} \\ -\sqrt{1/3} \end{pmatrix}$, $D = \begin{pmatrix} -\sqrt{1/3} \\ \sqrt{1/3} \end{pmatrix}$.

Now $f(0,0) = 0$, $f(A) = f(B) = 4$

$f(C) = f(D) = 15{,}3$. Hence the absolute minimum

is $(0,0)$, and there are two absolute maxima:

$C = \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \end{pmatrix}$, $D = \begin{pmatrix} -\sqrt{1/3} \\ \sqrt{1/3} \end{pmatrix}$

(d) min $f(x,y)$ subject to $x^2+y^2-xy \leq 1$
is a convex optimization problem because $f(x,y)$ is
a convex function ($H(f)$ is positive definite) and also
$g(x) = x^2+y^2-xy-1$ is a convex function ($H(g) = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ is positive definite) Hence by Theory, this is a convex optimization problem. Observe that $f(x,y)$ is a strongly convex function, because both eigenvalues are strictly positive and constant. Also $S = \{(x,y) \in \mathbb{R}^2 : g(x,y) \leq 0\}$ is convex and closed. Hence by a Theorem given in Theory, there exists a unique minimum of $f(x,y)$ in $S$.
Of course this is coherent with our previous computation in (c), because we obtained only one minimum $(0,0)$.
Note that, in contrast, there are two maxima, C and D.

② (a) $\int e^{-r^2} r\,dr = \frac{1}{2}\int e^{-t}\,dt = -\frac{1}{2}e^{-t} = -\frac{1}{2}e^{-r^2}$

$[t = r^2 \Rightarrow dt = 2r\,dr \Rightarrow r\,dr = \frac{1}{2}dt]$

Hence $\int_0^a e^{-r^2} r\,dr = \left[-\frac{1}{2}e^{-r^2}\right]_0^a = \frac{1}{2}\left(1-e^{-a^2}\right)$

(b) We have

$$\iint_{D_a} e^{-(x^2+y^2)}\,dx\,dy = \int_0^a \int_0^{2\pi} e^{-r^2} r\,d\theta\,dr =$$

$$= \int_0^{2\pi}\left(\int_0^a e^{-r^2} r\,dr\right)d\theta = (2\pi)\cdot\left[\frac{1}{2}\left(1-e^{-a^2}\right)\right] =$$

$$= \pi\left(1-e^{-a^2}\right) = \pi\left(1-\frac{1}{e^{a^2}}\right).$$

(c) We have

$$\iint_{S_a} e^{-(x^2+y^2)}\,dx\,dy = \int_{-a}^{a}\int_{-a}^{a} e^{-(x^2+y^2)}\,dx\,dy$$

$$= \int_{-a}^{a}\left(\int_{-a}^{a} e^{-y^2}\cdot e^{-x^2}\,dx\right)dy = \int_{-a}^{a} e^{-y^2}\left(\int_{-a}^{a} e^{-x^2}dx\right)dy$$

$$= \left(\int_{-a}^{a} e^{-x^2}dx\right)\left(\int_{-a}^{a} e^{-y^2}dy\right) = \left(\int_{-a}^{a} e^{-x^2}dx\right)^2.$$

Now since $D_a \subseteq S_a$ and $f(x,y) = e^{-(x^2+y^2)} > 0$,

we have $\iint_{D_a} f(x,y)\,dx\,dy \leq \iint_{S_a} f(x,y)\,dx\,dy$,

$$\begin{array}{ccc} \iint_{D_a} f(x,y)\,dx\,dy & \leq & \iint_{S_a} f(x,y)\,dx\,dy \\ \| & & \| \end{array}$$

which gives

$$\pi\left(1-\frac{1}{e^{a^2}}\right) = \iint_{D_a} e^{-(x^2+y^2)}\,dx\,dy \leq \left(\int_{-a}^{a} e^{-x^2}dx\right)^2.$$

(This can be used to show that $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$.)

<u>Short questions</u>

(a) $r'(t) = (t, \sqrt{t}, te^t) \Rightarrow r(t) = \left(\frac{t^2}{2}+A, \frac{2t^{3/2}}{3}+B, te^t-e^t+C\right)$

for constants $A, B, C$. Since $r(0) = (1,2,3)$ we get

$$r(t) = \left(\frac{t^2}{2}+1, \frac{2}{3}t^{3/2}+2, te^t-e^t+4\right).$$

(b) We compute $\nabla f = \begin{bmatrix} \ln(x_1)+1 \\ \ln(x_2)+1 \\ \vdots \\ \ln(x_n)+1 \end{bmatrix}$

and $H(f) = \begin{bmatrix} 1/x_1 & & & 0 \\ & 1/x_2 & & \\ & & \ddots & \\ 0 & & & 1/x_n \end{bmatrix}$.

Since all eigenvalues are positive (because $x_i > 0\ \forall i$) we conclude that $f(x_1, \dots, x_n)$ is a convex function.